

Signals

1.Signal.c

```
tejaswini@LAPTOP-S8US186U × + v
GNU nano 8.7.1 signal.c
#include<stdio.h>
#include<signal.h>
#include<unistd.h>
void handler(int sig)
{
printf("\nSIGINT received!\n");
printf("Signal number = %d\n", sig);
}
int main()
{
signal(SIGINT, handler);
while(1)
{
printf("Program is running...\n");
sleep(2);
}
return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc signal.c -o signal
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./signal
Program is running...
Program is running...
Program is running...
Program is running...
Program is running...
^C
SIGINT received!
Signal number = 2
Program is running...
Program is running...
Program is running...
^Z
[3]+  Stopped                  ./signal
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ |
```

2.Sigterm

```
tejaswini@LAPTOP-S8US186U x + v
GNU nano 8.7.1 sigterm.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
void handler(int sig)
{
if(sig==SIGTERM)
    printf("\nSIGTERM received!\n");
}
int main()
{
    signal(SIGTERM, handler);
    printf("Process ID: %d\n",getpid());
    printf("Waiting for SIGTERM...\n");
    while(1)
    {
        sleep(2);
    }
    return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U x + v
tejaswini@LAPTOP-S8US186U:~$ cd OS_Lab
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ kill -SIGTERM 660
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ |

fifo_write.c myfifo ques_1.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ nano sigterm.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc sigterm.c -o sigterm
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./sigterm
Process ID: 660
Waiting for SIGTERM...

SIGTERM received!
^Z
[4]+ Stopped ./sigterm
tejaswini@LAPTOP-S8US186U:~/OS_Lab$
```

3.Sigchild

```
tejaswini@LAPTOP-S8US186U × + v
GNU nano 8.7.1 sigchild.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>
#include <sys/wait.h>
void child_handler(int sig)
{
    printf("SIGCHLD received: Child has terminated\n");
    wait(NULL);
}
int main()
{
    signal(SIGCHLD, child_handler);

    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child is running...\n");
        sleep(2);
        printf("Child is exiting...\n");
    }
    else
    {
        printf("Parent is waiting...\n");
        sleep(5);
        printf("Parent finished.\n");
    }

    return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ nano sigchild.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc sigchild.c -o sigchild
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./sigchild
Parent is waiting...
Child is running...
Child is exiting...
SIGCHLD received: Child has terminated
Parent finished.
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ |
```

4.Sigpipe

```
tejaswini@LAPTOP-S8US186U × + v
GNU nano 8.7.1 sigpipe.c *
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
#include <stdlib.h>
void handle_sigpipe(int sig)
{
    printf("\nSIGPIPE received!\n");
    printf("The reading end of the pipe is closed.\n");
    exit(1);
}
int main()
{
    int fd[2];
    signal(SIGPIPE, handle_sigpipe);
    pipe(fd);
    printf("Pipe created.\n");
    // Close the reading end
    close(fd[0]);
    printf("Reading end closed.\n");
    printf("Trying to write to the pipe...\n");
    write(fd[1], "Hello", 5);
    printf("Write completed.\n");
    return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ nano sigpipe.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc sigpipe.c -o sigpipe
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./sigpipe
Pipe created.
Reading end closed.
Trying to write to the pipe...

SIGPIPE received!
The reading end of the pipe is closed.
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ |
```

5.Sigsegv

```
tejaswini@LAPTOP-S8US186U x + v
GNU nano 8.7.1 sigsegv.c *
#include <stdio.h>
#include <signal.h>
#include <stdlib.h>
void handle_sigsegv(int sig)
{
    printf("\nSIGSEGV received!\n");
    printf("Invalid memory access detected.\n");
    exit(1);
}
int main()
{
    signal(SIGSEGV, handle_sigsegv);
    printf("Program started.\n");
    int *p = NULL;
    printf("Trying to access invalid memory...\n");
    *p = 10;
    printf("Program completed.\n");
    return 0;
}
```

Output:

```
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ nano sigsegv.c
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ gcc sigsegv.c -o sigsegv
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ ./sigsegv
Program started.
Trying to access invalid memory...

SIGSEGV received!
Invalid memory access detected.
tejaswini@LAPTOP-S8US186U:~/OS_Lab$ |
```